

FROM ATOMIC ENTITIES TO PHYSICO-CHEMICAL SYSTEMS

The central theme of physical chemistry is the description of the behaviour of electrons within the atom. Understanding this behaviour is essential to understanding the behaviour of molecules (and, more generally, matter) from physical and chemical perspectives, i.e. their properties and reactivity.

1. CLASSICAL DESCRIPTION OF THE ATOM

1.1. Imaging atoms

The Greek philosopher Democritus (5th century BC) believed that matter could not be divided into smaller and smaller parts indefinitely. He thought that there must be fundamental building blocks, called **atoms**, which are supposed to be indivisible. In 1808, the English chemist Dalton argued that a given compound is always composed of the same ratio of different types of atom, each with a different mass.

This model was confirmed in the 20th century with the development of techniques such as scanning tunnelling microscopy (STM).

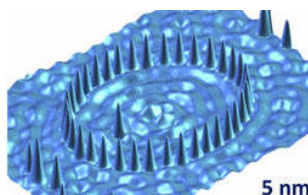


Figure 1-1 Elliptical quantum corral fabricated with cobalt atoms on the Cu surface using the CNST developed autonomous atom assembler (nist.gov)

The atom, which was initially considered a theoretical object, has become a real entity that can be visualised and manipulated.

1.2. Internal structure

Experiments conducted during the 19th century revealed the internal structure of the atom.

1.2.1. Electron

In 1897, the experiment of the English physicist Thomson (1906, NP in physics) led to the discovery of the electron.

A high voltage applied across two electrodes separated by a low-pressure gas generates a cathode ray. This cathode ray is deflected by an electric field.

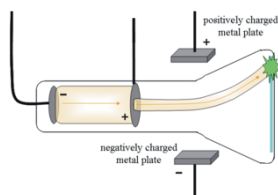


Figure 1-2 Cathode ray tube experiment (study.com)

1.2.2. Nucleus

In 1911, the experiment of the NZ physicist Rutherford led to the discovery of the nucleus.

A flow of α -particles (He^{2+}) is targeting a thin gold foil. Most of the α -particles passed through the foil, experiencing only minor deflections. However, a very small proportion was deflected significantly, with some even bouncing straight back.

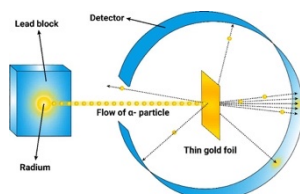


Figure 1-3 Thin gold foil bombarded with α -particles surrounded by a α -particles detector (vrlabacademy.com)

1.2.3. Properties of atoms and subatomic particles

The size of an atom is around 100 picometres ($1 \text{ pm} = 10^{-12} \text{ m}$), and the size of its nucleus is around 10^{-3} to 10^{-2} pm .

In 1909, the US physicist Millikan (1923, NP in physics) determined the charge of the electron. In the 1930s, experiments involving mass spectrometry of ions and photodisintegration of deuterium led to the determination of the masses of protons and neutrons.

entity	electron	proton	neutron	hydrogen	gold
charge (C)	-1.6×10^{-19}	$+1.6 \times 10^{-19}$	0	0	0
mass (kg)	9.1094×10^{-31}	1.6727×10^{-27}	1.6749×10^{-27}	1.6735×10^{-27}	3.2707×10^{-25}
atomic mass unit	1/1823	1.0073	1.0087	1.009	196.967

Table 1-1 Properties of atoms and subatomic particles

Atomic mass unit is defined as exactly 1/12th the mass of a single, unbound carbon-12 atom, at rest and in its ground state (see 2.5). $1 \text{ amu} = 1.6605 \times 10^{-27} \text{ kg}$

1.3. Chemical element and isotopes

The nucleus of an atom is represented by the **atomic number Z** and the **mass number A**. The corresponding atom has Z protons and (A-Z) neutrons in the nucleus as well as Z electrons surrounding the nucleus to ensure the atom's electrical neutrality.

A **chemical element** is defined by its atomic number Z. It is written as ${}^A_Z\text{X}$.

Isotopes are atomic nuclei that differ only in their mass number A.

2. QUANTUM MODEL OF THE ATOM

2.1. Electromagnetic radiation

ElectroMagnetic (EM) radiation is one of eight forms of **energy**:

Table 2-1 The 8 forms of energy

EM radiation can be described as oscillations of magnetic and electric fields. These oscillations travel like a wave at a propagation speed of $c = 3 \times 10^8 \text{ m} \cdot \text{s}^{-1}$ in a vacuum. Each EM radiation has a characteristic **temporal period** T and a spatial period known as the **wavelength** λ which are related by the equation:

$$\lambda = cT$$

The **frequency** ν of an EM radiation is given by the equation: $\nu = \frac{1}{T}$

The various types of EM radiation are displayed on the EM spectrum:

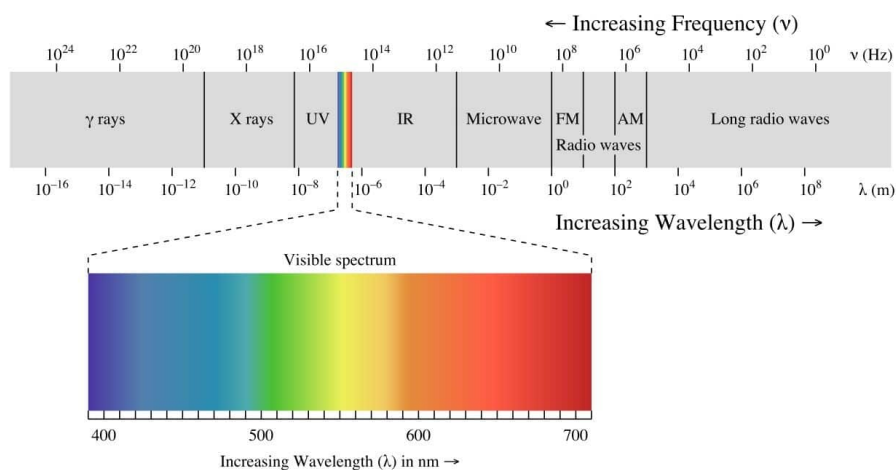


Figure 2-1 EM spectrum (leadertechnic.com)

2.2. Quantisation

EM radiation can also be described as a flux of **photons** which are particles that travel at a speed of c and are characterised by the energy formula of the German physicist Planck (1918, NP in physics):

$$E = h\nu$$

where h is Planck's constant $h = 6.62 \times 10^{-34} \text{ J} \cdot \text{s}$

The ability of EM radiation to be described as both a wave and a particle is known as: the wave-particle duality of EM radiation.

2.3. EM radiation-matter interaction

When it comes into contact with a matter, EM radiation interacts in several ways:

- Refraction & Reflection
- Scattering
- Absorption

When EM radiation passes through an aqueous solution of KMnO_4 , the **absorption** spectrum shows a broad, continuous black band.

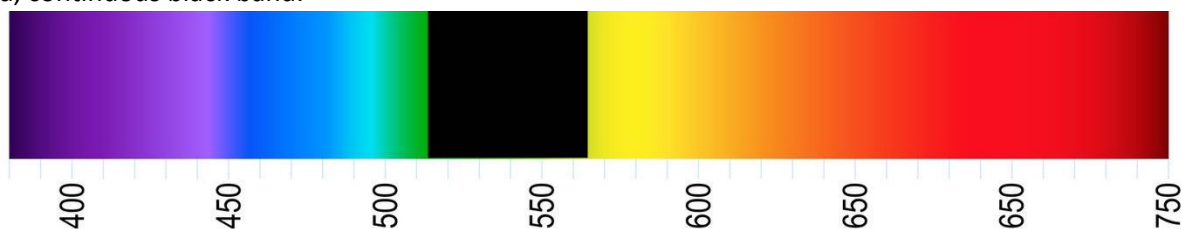


Figure 2-2 Absorption spectrum of $\text{KMnO}_4(\text{aq})$ (in nm)

2.4. Atomic spectra

2.4.1. Absorption spectrum

When EM radiation passes through a sample of hydrogen gas, the EM spectrum shows exactly four distinct black lines at wavelengths 656 nm, 486 nm, 434 nm and 410 nm.

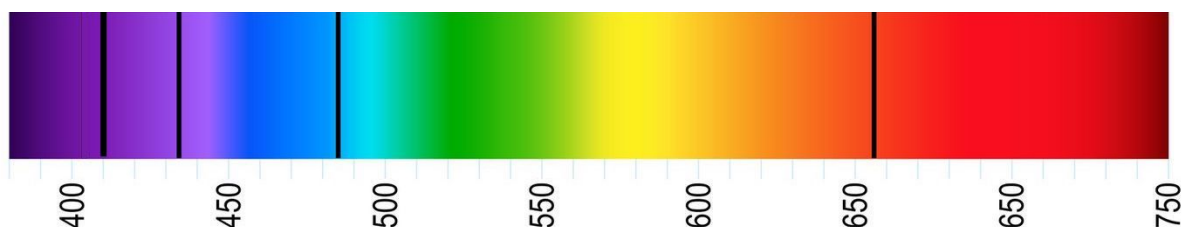


Figure 2-3 Absorption spectrum of hydrogen gas (in nm)

The experimental results show that hydrogen atoms can only absorb certain wavelengths of EM radiation. This suggests that atoms can only exchange discrete amounts of energy. This is known as the **quantised energy** of an atom.



2.4.2. Emission spectrum

When an electric discharge passes through a sample of hydrogen gas, EM radiation is produced by the gas. The **emission** spectrum shows exactly four distinct coloured lines at wavelengths 656 nm, 486 nm, 434 nm and 410 nm.

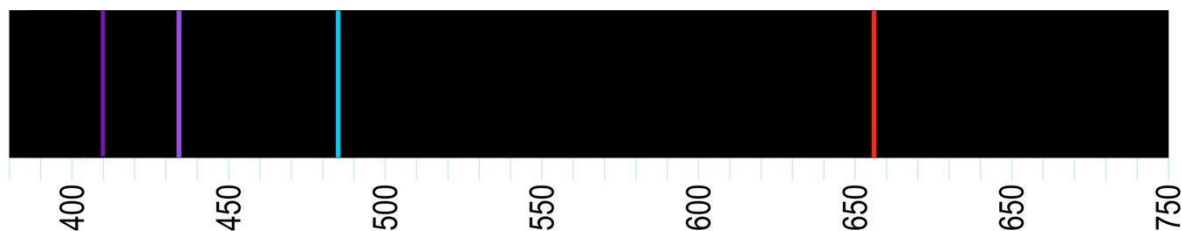


Figure 2-4 Emission spectrum of hydrogen gas (in nm)

2.4.3. Spectral series of hydrogen atom

In 1885, the Swiss mathematician Balmer identified the pattern of the hydrogen spectrum in the visible light:

$$\nu \propto \frac{1}{4} - \frac{1}{n^2}, n = 1, 2, 3 \dots$$

Other spectral series were discovered later in infrared and ultraviolet light. These experimental results led Rydberg (1888, Sweden) to generalise the equation to all spectral series:

$$\nu = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

where $(n_1, n_2) \in \mathbb{N}^*$, $n_1 < n_2$ and $R_H = 3.29 \times 10^{15} \text{ Hz}$

2.5. Electronic energy levels

In 1913, the Danish physicist Bohr (1922, NP in physics) put forward two postulates to explain the shape of the hydrogen EM spectrum:

- an electron bound to a proton can only assume certain specific energy levels:

$$E_n = -\frac{13.6}{n^2} \text{ (in eV)}, n \in \mathbb{N}^*. \text{ Its energy is **quantised**,$$

- the emission or absorption of EM radiation results from a **transition** of an electron between two energy levels.

For $n = 1$, the electron in the hydrogen atom has the lowest energy $E_1 = -13.6 \text{ eV}$, and the maximum interaction with the proton. This is known as the **ground state**.

For $n > 1$, the electron in the hydrogen atom has higher energy and weaker interaction with the proton. This is known as **excited state**.

For $n \rightarrow \infty$, $E_\infty = 0 \text{ eV}$ and the electron is no longer bonded to the proton. This corresponds to the **ionised state**.

During the emission process, an electron with energy E_p can return to a lower energy level E_n ($E_p > E_n$) by emitting a photon.	During the absorption process, an electron with energy E_n can be promoted to a higher energy level E_p ($E_p > E_n$) by absorbing a photon.

This photon is characterised by the quantum of energy: $\Delta E = E_p - E_n = h\nu = \frac{hc}{\lambda}$

3. QUANTUM NUMBERS AND ATOMIC ORBITALS

3.1. The nature of the electron

Wave-particle duality can be extended to electrons which can be described as both a particle (with mass and charge) and a wave (showing electronic diffraction).

In order to describe the electron in the hydrogen atom, we need to move away from the classical model of negatively charged particles orbiting the nucleus. Instead, the electron behaves like a wave, and we try to determine the probability of its presence in the space around the nucleus.

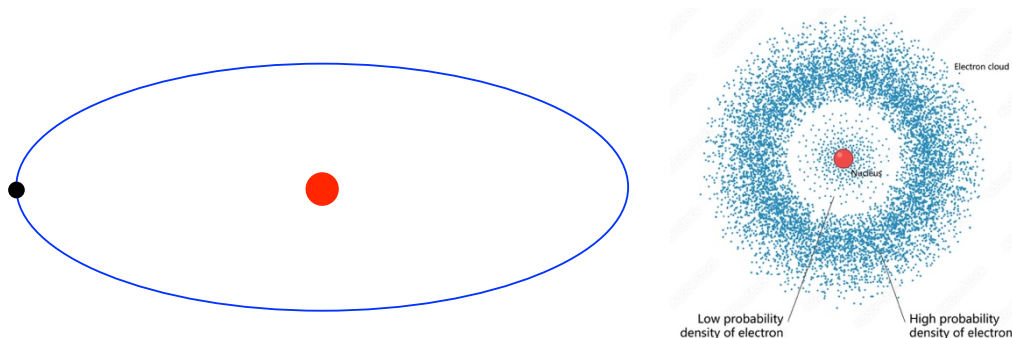


Figure 3-1 Classical and quantum description of the hydrogen atom (europosters.ie)

3.2. Hydrogen atom's electron and its quantum numbers n, ℓ, m_ℓ

Hydrogen is a single – electron atom. The quantum approach of this system is quite simple; three quantum numbers are sufficient to describe exactly the energy structure of the electron bound to the nucleus.

3.2.1. Principal quantum number n

The **principal quantum number n** is a natural number. It determines the electron shell in which the electron is located. According to Bohr's postulation, it is directly related to the electron's energy level:

$$E_n = -\frac{13.6}{n^2} \text{ (in eV)}, n \in \mathbb{N}^*$$

3.2.2. Secondary quantum number ℓ

The **secondary quantum number ℓ** is a whole number between 0 and $n - 1$. It determines the electron subshell in which the electron is located.

$$0 \leq \ell \leq n - 1, \ell \in \mathbb{N}$$

3.2.3. Magnetic quantum number m_ℓ

The **magnetic quantum number m_ℓ** is an integer between $-\ell$ and $+\ell$. It determines the spatial orientation of the electron subshell in which the electron is located.

$$-\ell \leq m_\ell \leq +\ell, m_\ell \in \mathbb{Z}$$

The electron subshells are oriented differently but have identical energy levels.

In 1896, the experiment of the Dutch physicist Zeeman (1902, NP in physics) revealed that the spectral lines split when an external magnetic field is applied to the atomic gas.



3.3. Hydrogen atom's electron and its atomic orbitals

The 3 quantum numbers n, ℓ, m_ℓ together define an **atomic orbital** (AO). This is a region around the nucleus in which the probability of finding the electron is given by the **wavefunction** Ψ_{n,ℓ,m_ℓ} .

n	E (in ev)	ℓ	m_ℓ	wavefunction Ψ_{n,ℓ,m_ℓ}	atomic orbital
1					
2					
3					

Table 3-1 Relationship between quantum numbers, wavefunctions and atomic orbitals

3.3.1. Energy levels of atomic orbitals

The energy level of AOs in a hydrogen atom depends only on the principal quantum number n : $E_n = -\frac{13.6}{n^2}$ (in eV). AOs with the same n value have the same energy level E_n : they are **degenerate**.

Figure 3-2 Diagram showing the energy levels of the hydrogen's AO
(one stroke represents one AO).

Note: an energy level of $E = 0$ eV corresponds to the absence of interaction between the nucleus and the electron. For a hydrogen-like ion (1 nucleus and 1 electron), $E_n = -\frac{13.6 \times Z^2}{n^2}$ (in eV).

3.3.2. Occupied and unoccupied atomic orbitals

An AO is occupied when the electron's behaviour is described by the corresponding wavefunction Ψ_{n,ℓ,m_ℓ} .

Example: The electron described by the wavefunction $\Psi_{2,1,1}$, occupies the 2p AO.

Ψ_{n,ℓ,m_ℓ}	1s	2p	3d	state
$\Psi_{1,0,0}$				
$\Psi_{3,2,-1}$				

Table 3-2 Occupied and unoccupied AO in the ground and excited state

3.3.3. Shapes of atomic orbitals

The shapes of AO are plotted using **boundary surfaces**. These surfaces contain a certain proportion of the probability of finding the electron around the nucleus, typically 95%.

In order to plot the shape of AO, it is necessary to determine the radial and angular **probability density** functions.

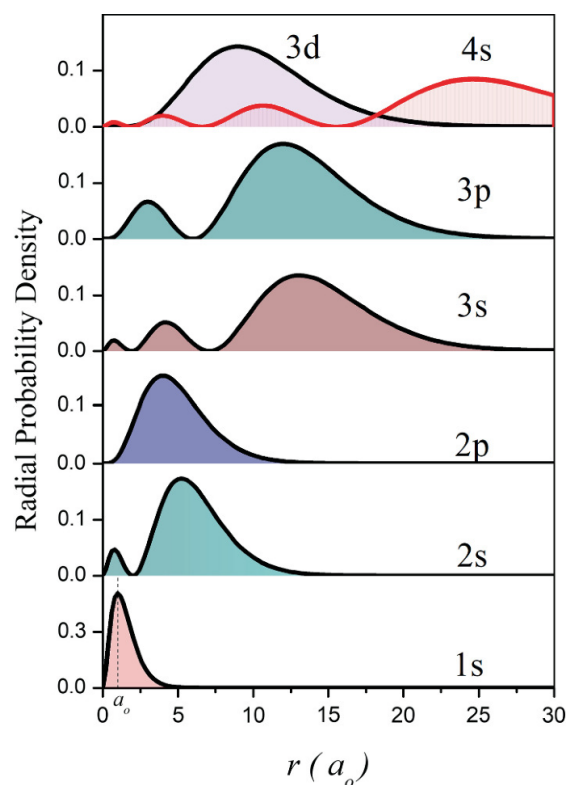


Figure 3-3 Radial probability density
(Joseyphus and Greneche, Introduction to Quantum Mechanics)

The **Bohr radius** (symbolized as a_0) is a fundamental physical constant that represents the most probable distance between the nucleus and the electron in a hydrogen atom in its ground state.

$$a_0 = 52,9 \text{ pm}$$

The AO radius of the hydrogen atoms and hydrogen-like ions is given by the equation: $r = \frac{n^2}{Z} a_0$

The radii at which the probability density is zero are called nodes. Every time a node is crossed, the wavefunction has a **phase shift** of an angle of π [2π]. This phase shift is indicated by using different colours or +/- labels on the AO shape.

The boundary surface for s-type AOs is spherical. As n increased, the AO gets larger:

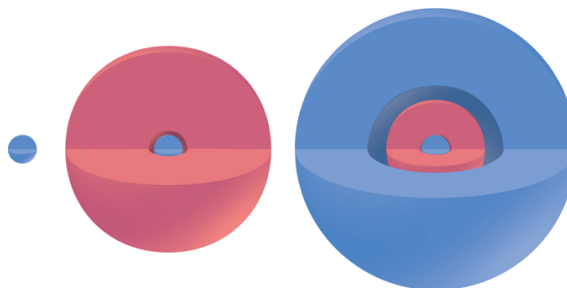


Figure 3-4 Shapes of s-type AOs (openstax.org)

The boundary surface for p-type AOs is a diaboloid shape but differs in their orientations:

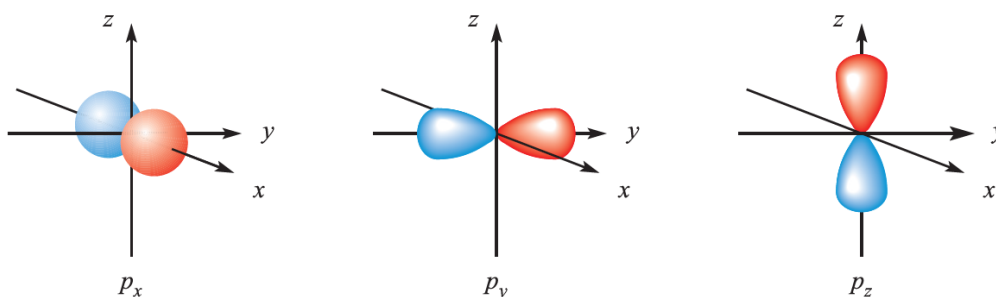


Figure 3-5 Shapes of p-type AOs
(Housecroft and Sharpe, Inorganic Chemistry)

AO	m_ℓ	(xOy)	(xOz)	(yOz)	Spatial orientation
p_x					
p_y					
p_z					

Table 3-3 Spatial orientation of p-type AOs

3.4. Many – electron atoms

Many-electron atoms have more than one electron. The quantum approach of this system is more complex due to the interactions between the electrons. Exact solutions to these systems cannot be calculated exactly (for now), but they can be approximated.

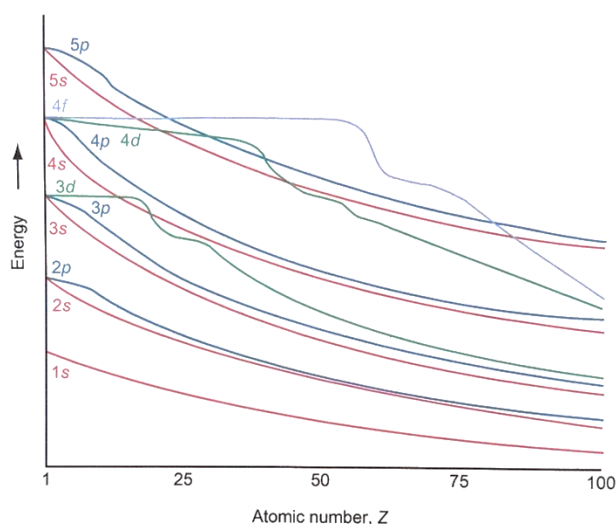
3.4.1. Orbital approximation

Electron – electron interactions are considered to be negligible compared with the nucleus – electron interaction. The wavefunction for a N – electron atom can be simplified as the product of N single – electron wavefunctions:

$$\psi(1,2, \dots, N) = \psi(1)\psi(2) \dots \psi(N)$$

Electrons occupy the AOs independently.

3.4.2. Energy levels of atomic orbitals

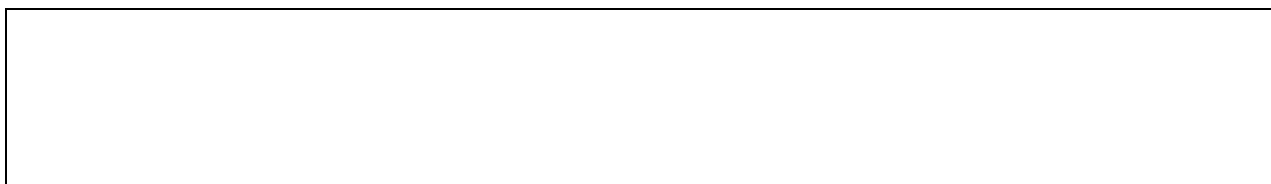


The energy level of AOs in a many – electron atom depends on n , ℓ and Z .

Figure 3-6 Variation of the AO energy levels
for many – electron atoms
(Burrows and Holman, Chemistry³)

Madelung's rule enables estimation of the energy level independently of Z:

1. The energy level of the OA increases as the sum of $n + \ell$ increases,
2. For an identical sum of $n + \ell$, the energy level increases as n increases.



3.4.3. Spin magnetic quantum number m_s

The **spin magnetic quantum number** m_s can only take on the following values $m_s = +\frac{1}{2}$ and $m_s = -\frac{1}{2}$. It represents the direction of the electron's spin which can be clockwise or anticlockwise, and is often represented by the symbol $\uparrow$ and $\downarrow$ respectively.

The emission spectrum of sodium (Na) shows a split line known as the D line. Instead of observing a single line, two lines very close in wavelength ($\lambda = 589.0$ nm and $\lambda = 589.6$ nm) are observed in the spectrum.